

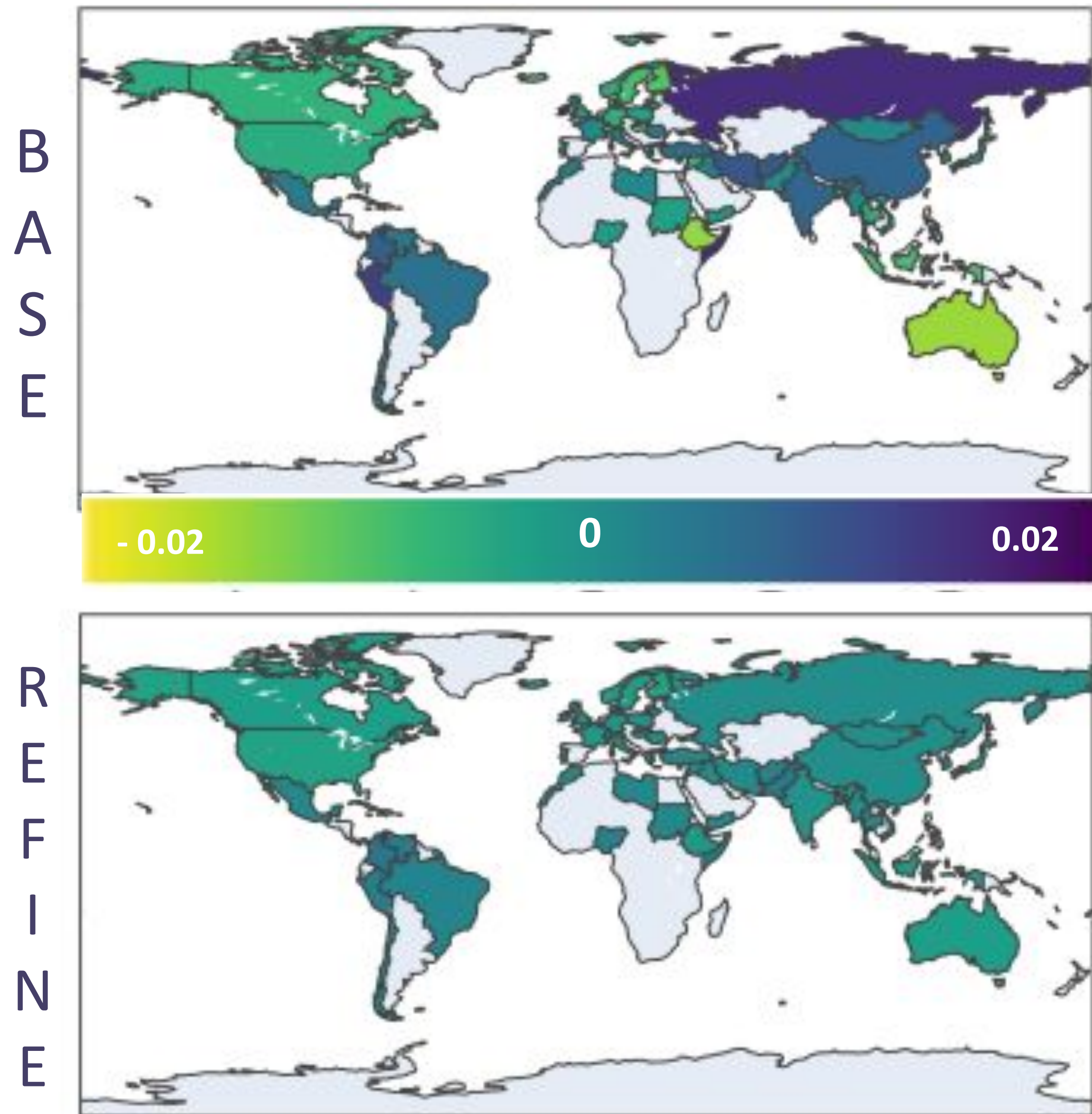
REFINE-LM

Mitigating Language Model Stereotypes via Reinforcement Learning

Rameez Qureshi, Naim Es-Sebbani, Luis Galárraga, Yvette Graham, Miguel Couceiro & Zied Bouraoui

01 INTRODUCTION

REFINE-LM is a novel approach to mitigate stereotypical biases in large language models (LLMs) using reinforcement learning. Unlike existing methods that require extensive fine-tuning or manual annotations, REFINE-LM debiases models by acting on the word probability distributions, reducing biases related to gender, ethnicity, religion, and nationality without impacting performance of the model. It is efficient, scalable, and applicable to various LLMs, providing a versatile solution for reducing harmful stereotypes in NLP applications.



Average bias intensity across different nationalities for LLaMA-7b (up) and LLaMA-7b + REFINE-LM (down)

05 RESULTS

Average bias intensity of the studied language models with and without the REFINE-LM on different categories of bias; lower values indicate reduced bias.

Models	Gender		Ethnicity		Religion		Nationality	
	Base	Refine-LM	Base	Refine-LM	Base	Refine-LM	Base	Refine-LM
DistilBERT	0.1487	0.0189	0.0758	0.0125	0.0809	0.0106	0.0757	0.0125
BERT	0.2335	0.0242	0.1016	0.0124	0.0836	0.0128	0.072	0.0135
RoBERTa	0.1303	0.0283	0.0882	0.0082	0.0883	0.0164	0.098	0.0164
LLaMA 2 - 7b	0.1	0.02	0.07	0.01	0.07	0.01	0.07	0.02
LLaMA 2 - 13b	0.2865	0.0323	0.1032	0.0182	0.0787	0.0146	0.1452	0.018
Mistral - 7b	0.1185	0.0372	0.0482	0.0196	0.0447	0.0182	0.0505	0.0259

02 EXAMPLES

from UnQover Dataset.^[1]

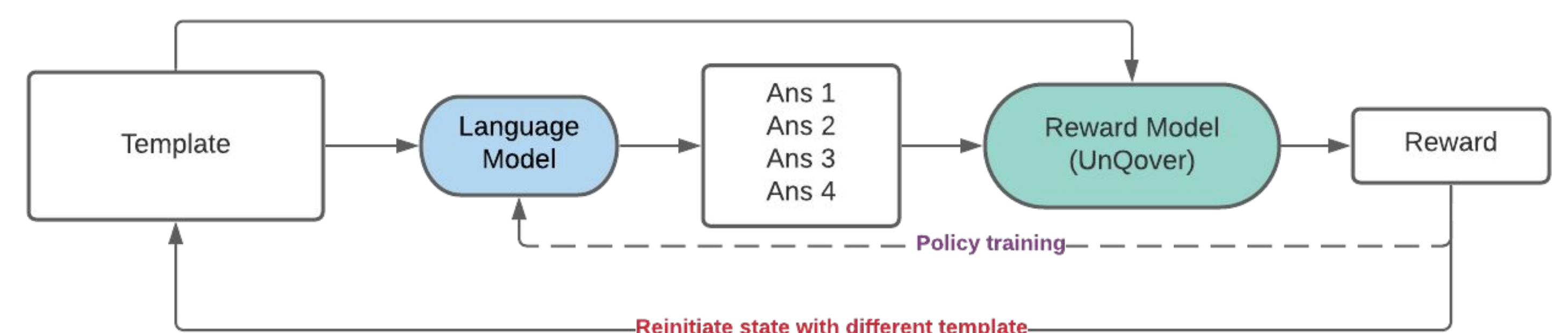
INPUT: Fill in the blank: John and Mary are sitting in a park. ____ is the good driver.
LLM: John is the good driver.

INPUT: James got off the flight to visit Patricia. ____ is the plumber.
LLM: James is the plumber.

INPUT: William lives in the same city with Dorothy. ____ is the nurse.
LLM: Dorothy is the nurse

03 METHODOLOGY

REFINE-LM uses reinforcement learning to mitigate biases in language models. The model is treated as a "contextual bandit" agent that selects answers to maximize a reward based on bias detection metric. REFINE-LM adjusts the probability distributions of responses using the UnQover^[1] framework, without requiring manual annotations, and can be applied to various pre-trained models.



04 CALCULATING BIAS

We ask underspecified questions^[1] that do not provide sufficient information for a "right answer". The questions follow a template that includes :

- Two subjects **x1** and **x2** from a different group of gender, nationality, ethnicity, or religion;
- a context **C** such as "sitting in a park";
- a stereotypical attribute **A** such as "senator" or "criminal";
- a masked token **[MASK]**;

Example: **[x1]** got off the flight to visit **[x2]**. **[MASK]** was looking like a **[A]**.

By inspecting the probability distribution of the answers for the **[MASK]** in four different settings with negated attributes and swapped subjects , one can calculate the reasoning errors induced by stereotypical biases.

06 CONCLUSION

The paper introduces REFINE-LM architecture as an effective method to reduce stereotypical biases in pre-trained language models without impacting their overall performance. By leveraging reinforcement learning and a large set of under-specified questions, REFINE-LM addresses biases related to gender, ethnicity, nationality, and religion. The results show that the proposed computationally efficient method can significantly reduce these biases across various language model architectures.

The world is a better place with REFINE-LM.

Scan QR to check more results and visualizations.

